ANERI BHAVSAR

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**AI/ML Engineer** with experience in Computer Vision, Deep Learning, and RAG systems. Skilled in Python, SQL, PyTorch, TensorFlow, OpenCV, and AWS. Experienced in developing real-time AI solutions for surveillance, industrial inspection, object detection, segmentation, and intelligent document systems.

## EDUCATION

**Bachelor of Technology: Computer Science-BDA** **2021-2025**

Course: Specialization in Big Data Analytics, 8.59

**Higher Secondary Education (HSC)** **2021**

Kendriya Vidyalaya Sabarmati, 88%

**Senior Secondary Education (SSC)** **2019**

Kendriya Vidyalaya Sabarmati, 85%

## TOOLS & TECHNOLOGIES

Python, SQL, PyTorch, TensorFlow, OpenCV, Computer Vision, Deep Learning, CNNs, Object Detection, Image Segmentation, Transfer Learning, Model Deployment, Facial Recognition, ANPR, Thermal Imaging, RAG, LangChain, LangGraph, LLMs, VLMs, Vector Databases, Ollama, Real-Time Inference, Edge AI, AWS, QGIS, ArcGIS

## EXPERIENCE

**Maharshi Industries Pvt.Ltd** **June 2025 –Present**
**AI/MLEngineer**

- **AI-Based Remote Quality Inspection & Monitoring System**
  - Developed an AI-powered quality inspection solution using **RF-DETR**, OpenCV, and PyTorch for automated defect detection and analysis.
  - Integrated **Vision-Language Models (Qwen,Gemma4 VLM)** for intelligent defect interpretation and reporting.
  - Designed real-time, low-latency inference pipelines for industrial camera streams..
- **AI Surveillance & Security System – Indian Army**
  - Built real-time surveillance solutions for human, vehicle, animal, and helicopter,BFSR detection using RGB and thermal cameras.using RF-DETR pipeline.
  - Developed Facial Recognition and ANPR systems for personnel identification, access control, and blacklist monitoring.
  - Optimized multi-camera RTSP-based inference pipelines for deployment in mission-critical environments.
- **Color Change Detection System – DRDE Gwalior (Defense Research Project)**
  - Developed a computer vision system for automated chemical resistance testing of military protective clothing.
  - Performed **camera calibration and image correction** to ensure accurate and consistent color measurement.
  - Implemented real-time color change detection using OpenCV and Deep Learning.
  - Integrated AI models with robotic inspection workflows for remote and contactless testing.

**BISAG-N** **January 2025 – April 2025**

**Data Science Intern**

Worked on developing and optimizing deep learning models for geospatial AI applications, focusing on object detection and segmentation. Gained hands-on experience with several deep learning models, leveraging Python, TensorFlow, PyTorch, and OpenCV for model training and evaluation. Applied key metrics like Accuracy, Precision, Recall, IoU, and mAP to assess performance while utilizing QGIS, SNAP Desktop, and Roboflow for geospatial data processing.

## PROJECTS

**Tree Detection using Deep Learning Models and Perform Spatial Temporal Analysis:** Built a tree detection and instance segmentation tool using YOLOv8-seg, YOLOv11-seg, and Mask R-CNN on UAV and satellite imagery. Annotated data in Roboflow (COCO & YOLO formats) for exact tree mapping. QGIS was incorporated for geospatial analysis, also enabling the canopy-based tree counting.

**Rooftop Detection with Light Weight & Other Deep Learning Models on Satellite Imagery:** Developed a rooftop detection system using YOLOv8s-seg, YOLOv11s-seg, DeepLabV3+ (ResNet50), and U-Net (ResNet50) on satellite imagery. Annotated data in Roboflow (COCO & YOLO formats) for exact roof Segmentation. Integrated geospatial analysis to enhance mapping accuracy and support urban planning.

**Collaborative Filtering Recommendation System:** Developed a user-based collaborative filtering recommendation model, leveraging model training techniques to enhance personalized recommendations. Designed and implemented a user-friendly system to improve accessibility and usability.

**Ask ANERI:** Developed ASK ANERI, an AI-powered health and skincare chatbot using RAG, LangChain, and ChromaDB. Built a custom knowledge base covering acne, PCOS, hormonal health, skincare, and women's wellness, enabling context-aware retrieval and intelligent response generation through semantic search and LLM integration.

## CERTIFIATIONS

- Learn Python: The Complete Python Programming Course 2023 Udemy
- Introduction to Gen AI 2024 Google Cloud – badge
- Predictive Analytics using IBM spss modeler 2024 IBM
- Data Analysis With Python 2022 IBM
- AWS Academy Graduate - AWS Academy Cloud Foundations 2022